

TRExSIM ALPHA RESEARCH · US TOP1000 · 2007-2021

Three alphas, three mechanisms

Event-drift momentum, two-sided price reversion and an orthogonal two-leg composite, each benchmarked against the account's 80 submitted alphas.

361878 · Momentum

320000 · Mean reversion

336690 · Composite

All statistics are in-sample (Slow period) and in the platform's own units.

How every alpha is benchmarked

The test bed

Universe US TOP1000, industry-neutral

Period 15 Mar 2007 – 30 Dec 2021 (Slow)

Frequency Daily rebalance, delay 0 (trade the close)

Inputs Daily bars + daily summaries of intraday data (last-hour VWAP, volume share, skew)

Book Long-short, dollar-neutral: the right yardstick is peer alphas, not the S&P 500

1

Pool

Median and rank among all 80 submitted alphas

2

Family peers

Median of alphas built on the same mechanism

3

Construction

For the composite: each leg alone, and what the maths predicts for the pair

IR = mean $\div$ st. dev. of daily PnL. TVR = daily turnover. IR/ $\sqrt{\text{TVR}}$ = the platform score. Ret/DD = return $\div$ max drawdown.

0.102

Median IR

0.96

Median turnover (TVR)

0.118

Median IR/ $\sqrt{\text{TVR}}$

10.8

Median drawdown

Momentum: ride earnings reactions that volume confirmed

361878 • Momentum

1

Spot the event

Days-to-next-earnings resets by >10, or net income changes: a report just landed

2

Score the reaction

Return ÷ its 60-day volatility × volume ÷ its 60-day average

3

Hold it for 20 days, same direction

Under-reaction to earnings news: post-earnings drift (Bernard & Thomas, 1989)

```
flow = ts_sum(event × (ret1/ts_std(ret1,60))  
              × (volume/ts_mean(volume,60)), 20)  
alpha = cs_zscore(2 × cs_zscore(flow) +  
earnings_prox)
```

earnings_prox = cs_remove_middle(1/days_to_next_earnings, 0.5): a smaller pre-announcement risk premium

Metric	361878	Family C med.	Pool median	Rank / 80
IR	0.074 ▼	0.088	0.102	76
Turnover	0.368 ▲	0.984	0.958	14
IR/√TVR	0.121 ▲	0.099	0.118	34
Drawdown	22.4 ▼	11.7	10.8	76
Ret/DD	0.0027 ▼	0.0053	0.0062	79

Verdict: weak alone, valuable in the book

Lowest IR in its family and a deep drawdown, but it turns over less than a third as much as its peers, so it still beats the pool median score. Its sign is continuation, the opposite of every reversal signal on the account, which is what makes it worth holding next to them.

Mean reversion: get paid to absorb forced flow

320000 • Mean reversion

1

Displaced at the open

Open far from the 10-day VWAP: overnight order imbalance

2

Still stretched at the close

Close far from the day's VWAP: pressure, not news, if both legs agree

3

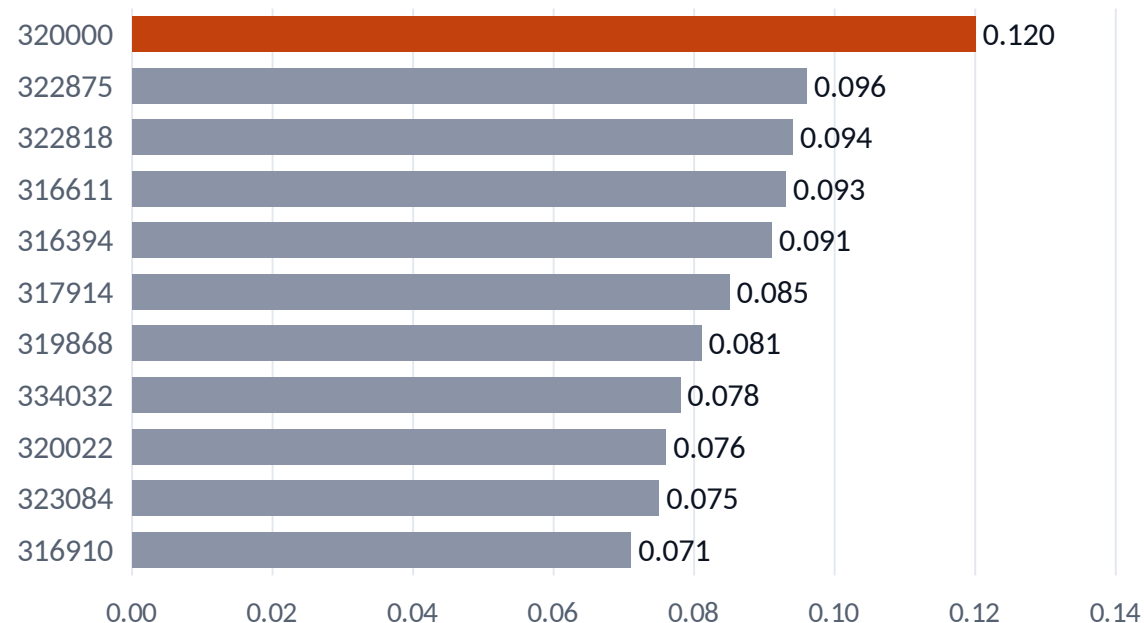
Take the other side for 1–5 days

The liquidity provider is paid as price normalises (Nagel, 2012)

```
alpha = -cs_rank(open - ts_mean(vwap, 10))  
        × abs(cs_rank(close - vwap))
```

Structure of Formulaic Alpha #5 (Kakushadze, 2016); the saved TrexSim expression is an adaptation.

IR of the 11 single intraday signals (Family A)



1 / 80

Return 0.145 (#1 on the account)

15 / 80

IR 0.120 (pool med. 0.102)

9 / 80

Ret/DD 0.011 (med. 0.006)

Composite: two weak, unrelated views of the close

336690 • Composite

Leg 1 • Who traded

vol-conc (alone: 323084)

`cs_zscore(volume_last_hour / volume)`

Benchmark-driven and institutional orders cluster at the close and are worked over several days, so their pressure persists.

Persists (+) IR 0.075 • rank 75 / 80

Leg 2 • Where it closed

day-range (alone: 316611)

`-cs_zscore((close - low) / (high - low))`

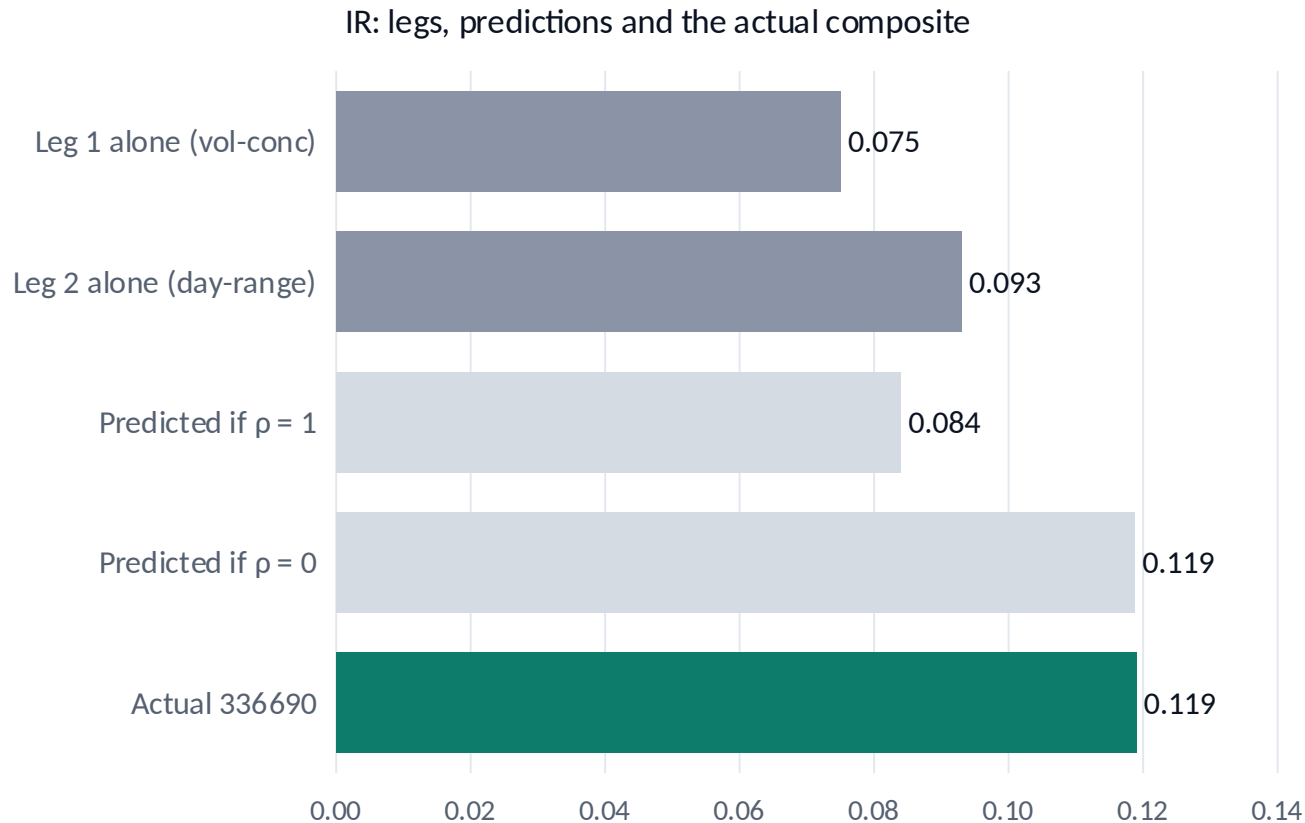
A close pinned at the high means buyers were still pushing at the bell. Supplying that liquidity pays as it fades.

Reverts (-) IR 0.093 • rank 52 / 80

`alpha 336690 = cs_zscore( vol_conc + day_range_reversal )` equal weights, no fitting

Why they are orthogonal: different input (volume share vs price location), different behaviour (persists vs reverts), different question (who vs where).

The numbers say the legs are independent



Equal-weight sum of two z-scored legs

$$IR = (IR_1 + IR_2) \div \sqrt{2 + 2\rho}$$

Solve for ρ with $IR = 0.119$:

$$\rho \approx 0.00$$

+59% IR over the weaker leg, +28% over the stronger

IR rank 16 / 80 and return rank 7 / 80, from two legs ranked 75 and 52.

The cost: turnover 1.37 (rank 74 / 80) leaves its score at 0.102, below the 0.118 pool median.

Assumes both legs have equal volatility after z-scoring. Leg IRs are from their own submitted runs (323084, 316611).

Scorecard against the 80-alpha pool

Metric	Pool median	Pool top quartile	361878 Momentum	320000 Reversion	336690 Composite
IR	0.102	0.118	0.074 ▼ 76 / 80	0.120 ▲ 15 / 80	0.119 ▲ 16 / 80
Return	0.065	0.094	0.060 ▼ 47 / 80	0.145 ▲ 1 / 80	0.108 ▲ 7 / 80
Turnover (lower is better)	0.958	0.58 or less	0.368 ▲ 14 / 80	0.987 ▼ 44 / 80	1.369 ▼ 74 / 80
IR/√TVR (platform score)	0.118	0.134	0.121 ▲ 34 / 80	0.121 ▲ 34 / 80	0.102 ▼ 54 / 80
Drawdown (lower is better)	10.8	7.6 or less	22.4 ▼ 76 / 80	13.0 ▼ 54 / 80	20.5 ▼ 71 / 80
Ret/DD	0.0062	0.0087	0.0027 ▼ 79 / 80	0.0112 ▲ 9 / 80	0.0053 ▼ 51 / 80

Momentum

Cheapest to trade, weakest alone. Earns its place by betting the opposite way to the reversal pool.

Reversion

Strongest standalone: #1 return on the account and top-decile Ret/DD. Turnover is its cost.

Composite

Proof of construction: two bottom-half legs make a top-20 IR. Turnover and drawdown are the price.

What the three add up to

361878 • Momentum

Horizon ~20 days • bets continuation

Fails if post-earnings drift gets arbitrated away, or on earnings seasons where reactions reverse (max DD 22.4).

320000 • Mean reversion

Horizon 1-5 days • bets reversal

Fails when a move is news rather than pressure, and in liquidity crises when providers are run over.

336690 • Composite

Horizon 1-5 days • persist + revert

Fails if late volume and close location start moving together, so ρ rises and the gain over the legs shrinks.

Next checks before trusting this out of sample

Measure the pairwise correlation of the three PnL streams • run them on out-of-sample dates • check drawdowns by year

All figures are in-sample backtests on TrexSim; the hypotheses explain why the signals might work, they are not proven causes.